

Researcher Yang Ya from Beijing Institute of Nanoenergy and Nanosystems and Professor Jun Chen from UCLA introduced a self-powered flexible intelligent keyboard (SFIK), which uses huge magneto-elastic effect to convert mechanical pressure generated by keys into electrical signals. The sensor has a wide trigger sensing distance (35 to 600 kPa) and a fast response time (~ 300 ms), and can record and identify the dynamics of a single keystroke. The keyboard is integrated with machine learning, and supports authentication through fixed text and dynamic text input. It accurately verifies the 8-character fixed password with a success rate of 95.3%, and verifies the dynamic text from 14 groups of double keys with 100% accuracy. In view of its functions, SFIK provides promising applications in artificial intelligence, network security and access control of computers and networks.

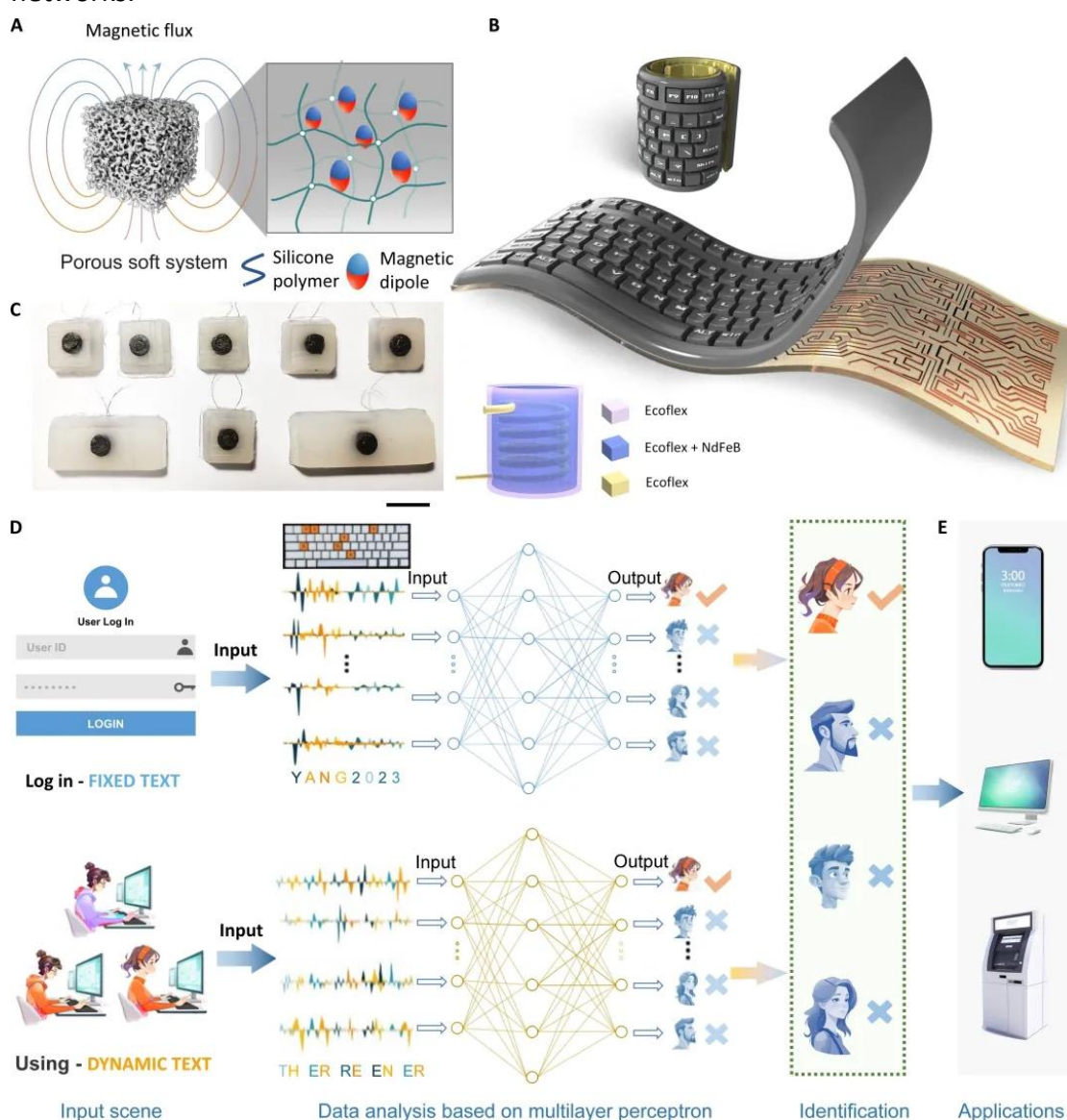


Figure 1. Overview of self-powered flexible intelligent keyboard for personalized keystroke dynamics.

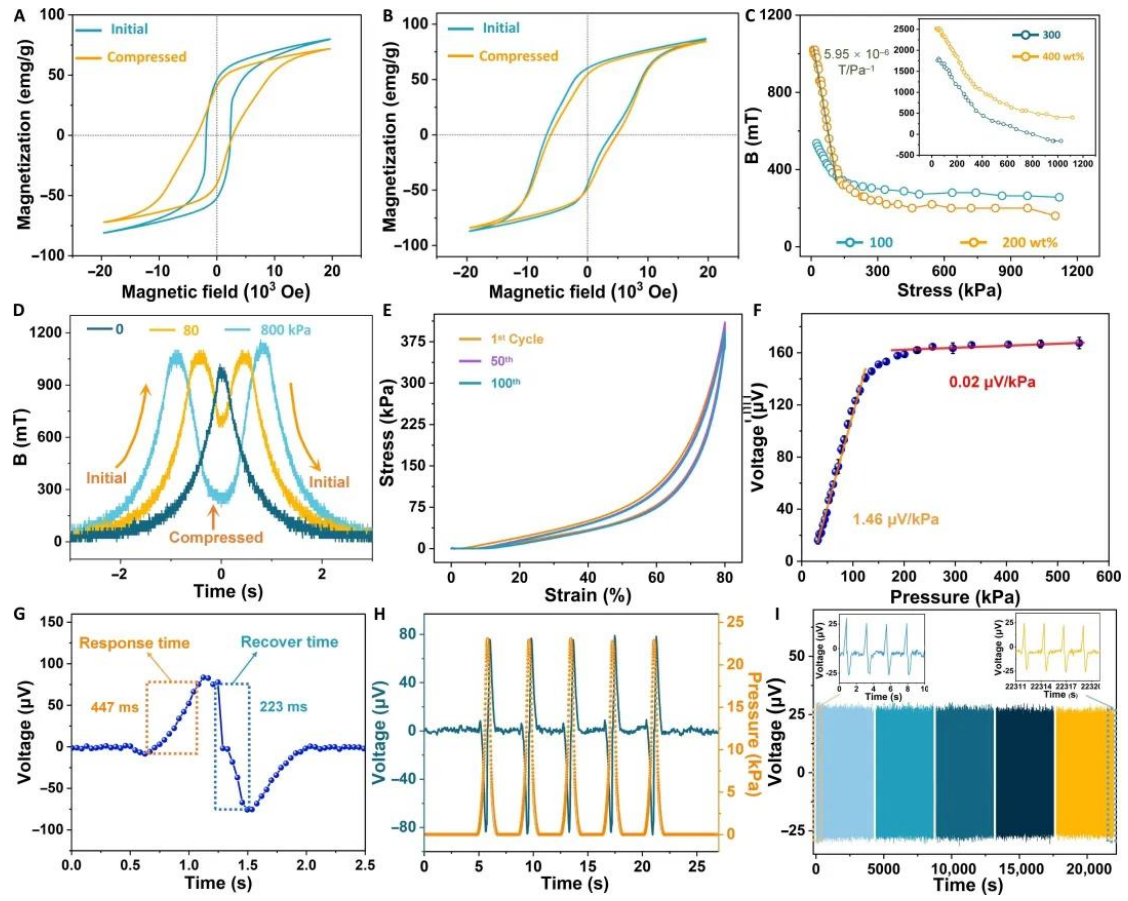


Figure 2. Pressure sensing evaluation of each magnetoelastic key unit.

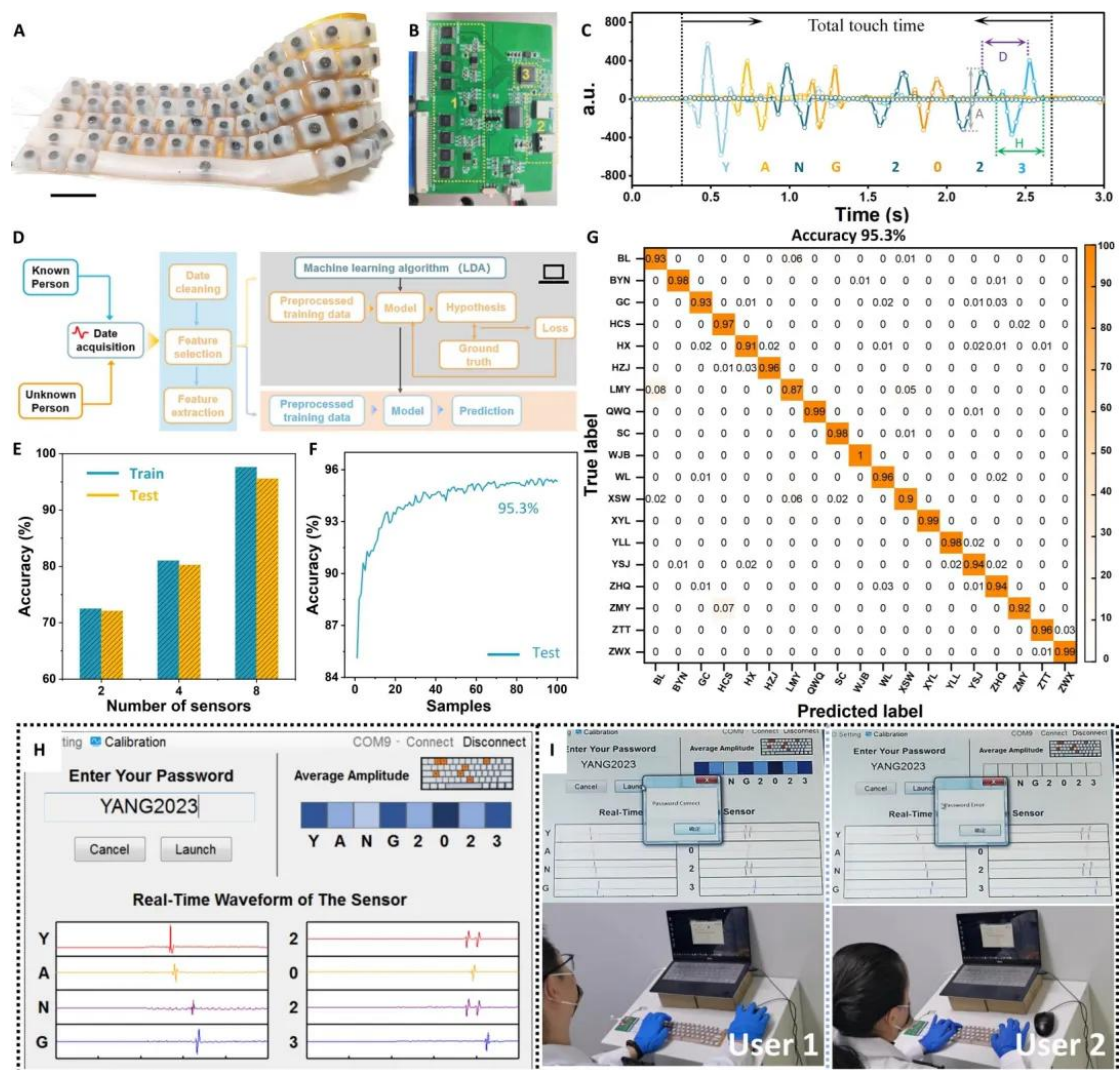


Figure 3. Multidimensional identity authentication system based on magneto-elastic smart keyboard.

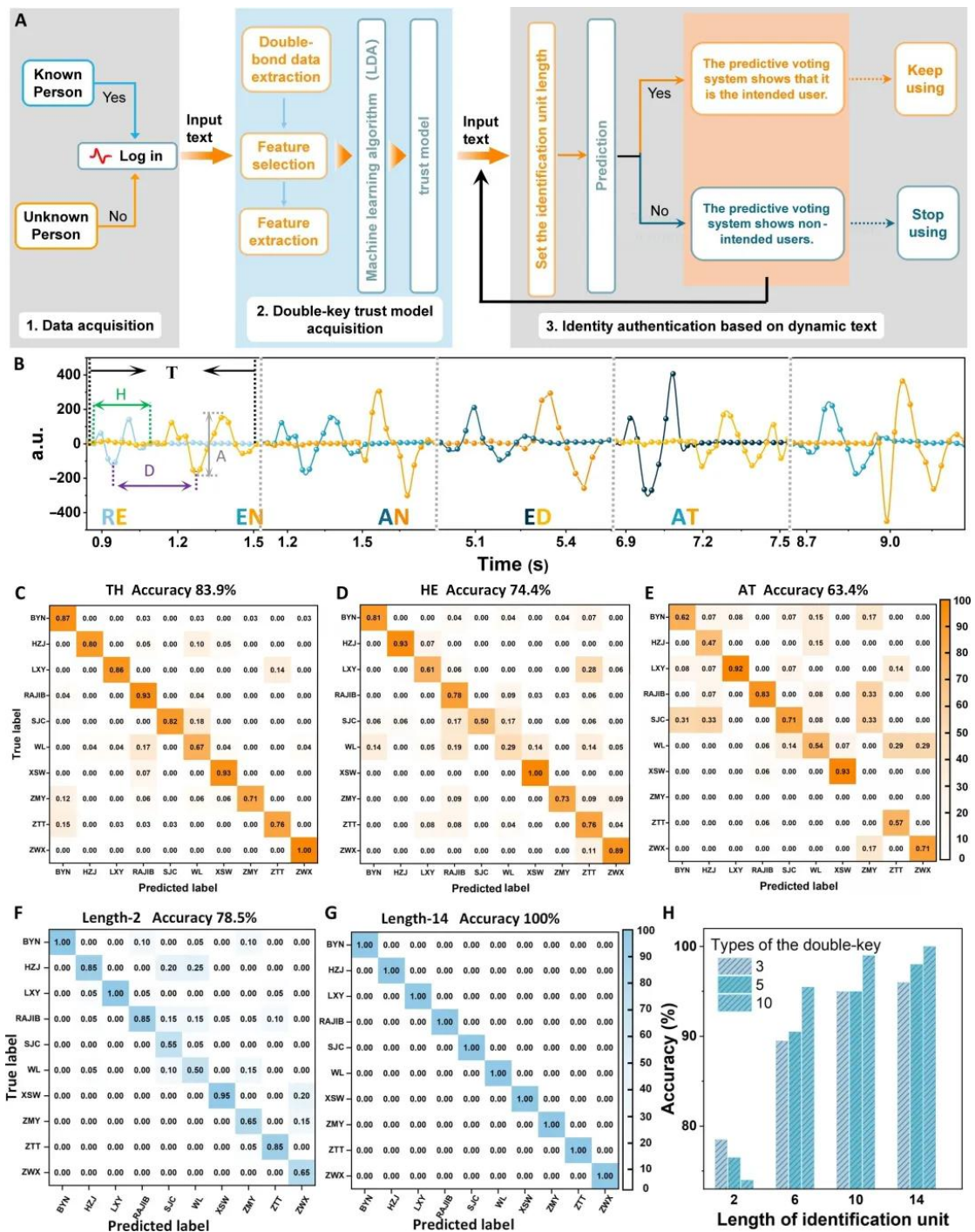


Figure 4. Personalized keystroke dynamics to achieve multi-dimensional identity authentication.

Original title: A flexible pressure sensor array for self-powered identity Authentication during typing

Original link: <https://www.science.org/doi/10.1126/sciadv.ads2297#abstract>

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